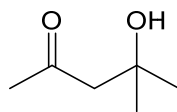
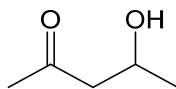
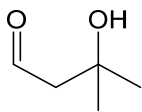
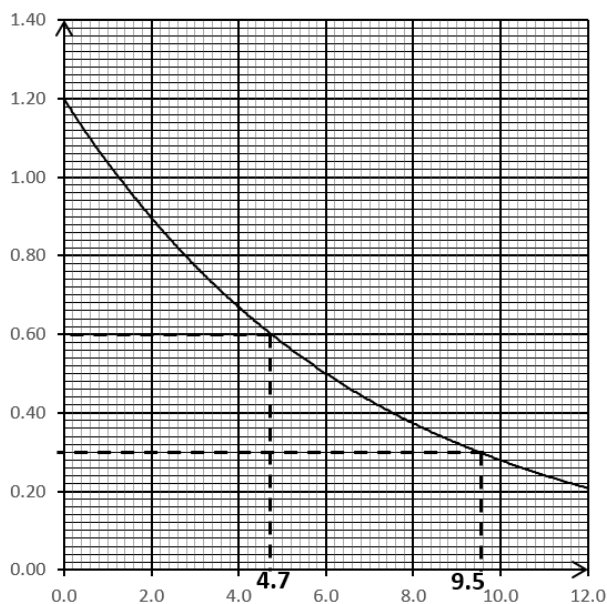

 2018 Y6 H2 Chemistry Preliminary Exams Paper 3 (Suggested Solutions)

Section A

1 (a) (i)



- (ii) Since OH^- is a catalyst, $[\text{OH}^-]$ is constant during each experiment and does not affect the reaction rate.



Using Experiment I,

First $t_{1/2} = 4.7 \text{ min}$. Second $t_{1/2} = 9.5 - 4.7 = 4.8 \text{ min}$.

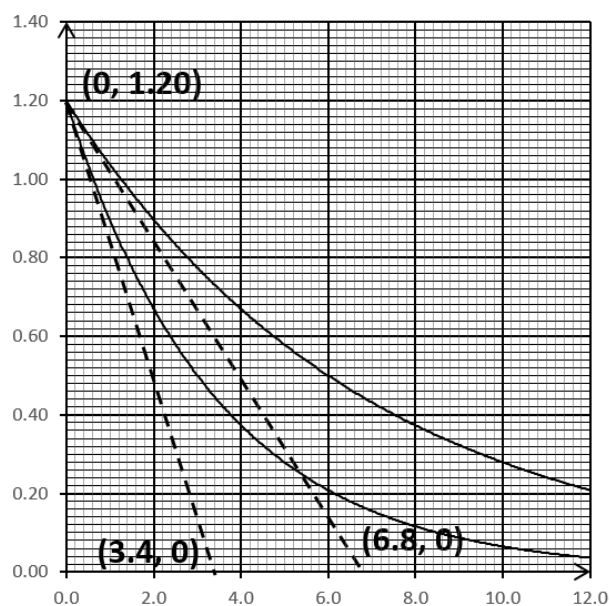
Since both $t_{1/2}$ are approximately the same, the reaction is first order w.r.t. ethanal.

- (iii) By drawing tangents at $t = 0 \text{ min}$,

$$\text{For Experiment I, initial rate} = \left| \frac{1.20 - 0}{0 - 6.8} \right| = \underline{0.176 \text{ mol dm}^{-3} \text{ min}^{-1}}$$

$$\text{For Experiment II, initial rate} = \left| \frac{1.20 - 0}{0 - 3.4} \right| = \underline{0.353 \text{ mol dm}^{-3} \text{ min}^{-1}}$$

When $[\text{OH}^-]$ doubles, rate doubles. Hence, reaction is first order w.r.t. OH^- .



(iv) $\text{rate} = k[\text{OH}^-][\text{ethanal}]$

(v) Initial rate method

Using Experiment I, $0.1765 = k(1.0)(1.2) \rightarrow k = \underline{0.147 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}}$

OR

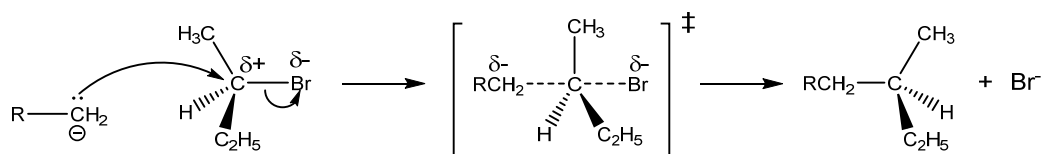
Half-life method

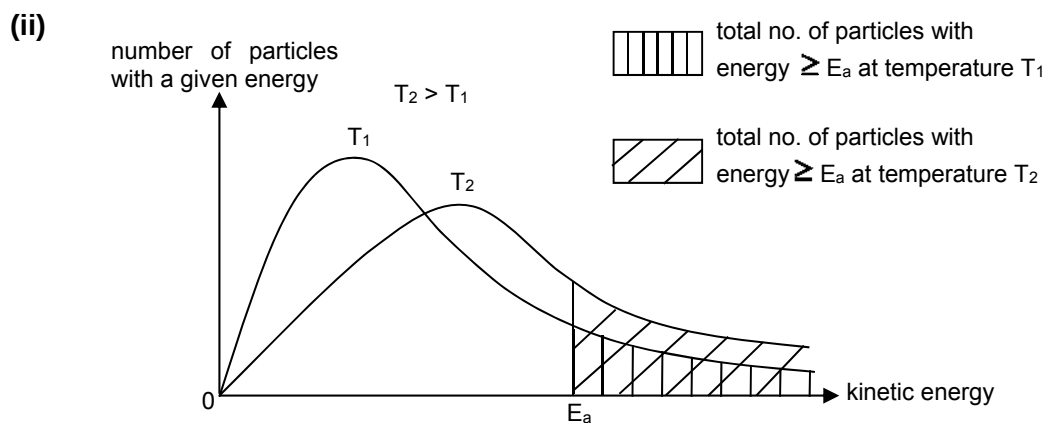
$$t_{1/2} = \frac{\ln 2}{k[\text{OH}^-]}$$

Using Experiment I, $4.75 = \frac{\ln 2}{k(1.0)} \rightarrow k = \underline{0.146 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}}$

(b) (i) Mechanism: S_N2 (or bimolecular nucleophilic substitution)

Let RCH_2^- represent A.





An increase in temperature from T_1 to T_2 increases the average kinetic energy of the reactant particles. More reactant particles have energy greater than or equal to the activation energy of the reaction.

This results in an increase in effective collision frequency and hence an increase in the rate of the reaction.

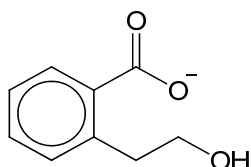
(iii) The C–Cl bond (BE = 340 kJ mol⁻¹) is stronger than the C–Br bond (BE = 280 kJ mol⁻¹). Hence it is more difficult to break the C–Cl bond, resulting in a slower reaction for 2-chlorobutane.

(iv) Hydroxide ions are less bulky and will experience less steric hindrance when attacking the electron deficient carbon atom.

OR

In **A**, the negative charge is delocalised into the adjacent C=O group. Hence, lone pair on A is less available, resulting in a slower reaction.

(c) (i)



(ii) Step 1: I₂(aq) with NaOH(aq), heat or warm
 Step 2: (catalytic) conc H₂SO₄, heat

2 (a) (i) PbF₂, PbCl₂ and PbBr₂ have giant ionic structure with strong ionic bonds.

PbF₂, PbCl₂ and PbBr₂ have the same cation and the anions have the same charge. Since the ionic radius increases from F⁻ to Br⁻, the interionic distance increases from PbF₂ to PbBr₂. Hence the strength of ionic bond and melting point decreases from PbF₂ to PbBr₂.

Or since $|LE| \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$ and the ionic radius increases from F⁻ to Br⁻, the LE

become less exothermic, less energy is required to overcome the ionic bond and melting point decreases from PbF₂ to PbBr₂.

- (ii) Due to its high charge and small size, Pb^{4+} has a high charge density and high polarising power, which causes it to distort the electron cloud of Cl^- to such an extent that PbCl_4 is predominantly a covalent compound.

Hence, PbCl_4 , having a simple covalent/molecular structure, consists of discrete molecules held together by weak instantaneous dipole-induced dipole interactions which requires a small amount of energy to overcome. Thus, it has a low melting point.

- (iii) $[\text{I}^-] = \frac{3.58 \times 10^{-5}}{25.0/1000} = 1.432 \times 10^{-3} \text{ mol dm}^{-3}$
- $$[\text{Pb}^{2+}] \text{ in filtrate} = \frac{9.80 \times 10^{-9}}{(1.431 \times 10^{-3})^2} = 4.779 \times 10^{-3} \text{ mol dm}^{-3}$$
- $$[\text{Cl}^-] \text{ from } \text{PbCl}_2 \text{ in filtrate} = 4.779 \times 10^{-3} \times 2 = 9.558 \times 10^{-3} \text{ mol dm}^{-3}$$
- $$\text{Total } [\text{Cl}^-] \text{ in filtrate} = 9.558 \times 10^{-3} + 0.050 = 0.05956 \text{ mol dm}^{-3}$$
- $$K_{\text{sp}} = [\text{Pb}^{2+}][\text{Cl}^-]^2 = (4.779 \times 10^{-3})(0.05956)^2 = \underline{1.70 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}}$$

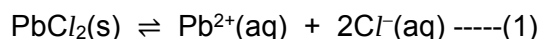
- (iv)
- | | | | |
|------|---------|----------|-----------|
| | H | Pb | Cl |
| mass | 0.6 | 59.0 | 40.4 |
| mole | 0.6/1.0 | 59.0/207 | 40.4/35.5 |
| | = 0.6 | = 0.285 | = 1.138 |
| | ≈ 2 | ≈ 1 | ≈ 4 |

Empirical formula = H_2PbCl_4

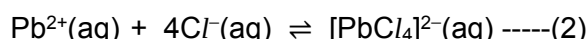
Actual formula = H_2PbCl_4

Complex ion = $[\text{PbCl}_4]^{2-}$

- (v) The following equilibrium is established in the saturated solution:



When concentrated hydrochloric acid is added and the mixture shaken, $[\text{Cl}^-]$ is greatly increased. Pb^{2+} ions react with Cl^- to form the soluble complex, $[\text{PbCl}_4]^{2-}$, as shown below.



The formation of $[\text{PbCl}_4]^{2-}$ decreases the $[\text{Pb}^{2+}]$ in the solution.

To counteract the decrease in $[\text{Pb}^{2+}]$, the equilibrium position of reaction (1) shifts to the right i.e. PbCl_2 dissolves.

or

$[\text{Pb}^{2+}]$ decreases due to the formation of $[\text{PbCl}_4]^{2-}$. When ionic product, $[\text{Pb}^{2+}][\text{Cl}^-]^2$ becomes less than the K_{sp} , all the PbCl_2 dissolves.

- (b) (i) $2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$

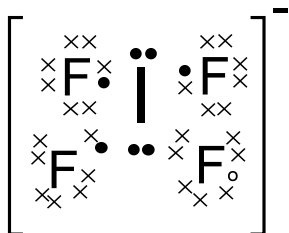
- (ii) The thermal stabilities of HCl to HI , which is related to the H-X bond strength,

increase in the order: $\text{HI} < \text{HBr} < \text{HCl}$

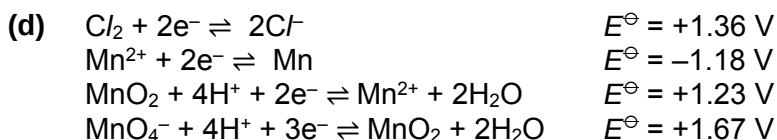
The stronger the H-X bond, the more thermally stable HX is. Since the bond strength of $\text{HI} < \text{HBr} < \text{HCl}$, the thermal stability of $\text{HI} < \text{HBr} < \text{HCl}$.

The energy supplied by the red hot wire was insufficient to decompose HCl , hence there is no observable change; was able to decompose some HBr , resulting in reddish-brown vapour of Br_2 formed slowly; and was able to decompose HI readily to result in purple fumes of I_2 evolved vigorously.

(c)



Square planar, 90°



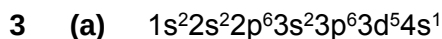
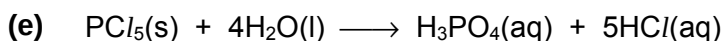
Considering the reaction between Cl_2 and Mn , $E^\ominus_{\text{cell}} = +1.36 - (-1.18)$
 $= +2.54 \text{ V (spontaneous)}$

Considering the reaction between Cl_2 and Mn^{2+} , $E^\ominus_{\text{cell}} = +1.36 - 1.23$
 $= +0.13 \text{ V (spontaneous)}$

Considering the reaction between Cl_2 and MnO_2 , $E^\ominus_{\text{cell}} = +1.36 - 1.67$
 $= -0.31 \text{ V (not spontaneous)}$

Thus, Cl_2 will oxidise Mn to Mn^{2+} and further oxidise Mn^{2+} to MnO_2 . (proven)

Overall eqn: $2\text{Cl}_2 + \text{Mn} + 2\text{H}_2\text{O} \longrightarrow 4\text{Cl}^- + \text{MnO}_2 + 4\text{H}^+$



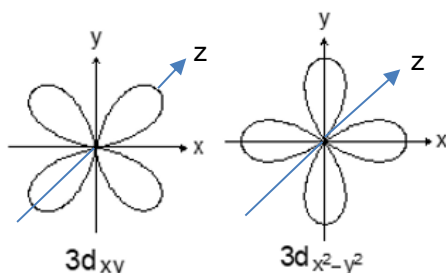
- (b) (i) Step I: acid-base (or complexation or complex formation)
 Step II: oxidation / redox reaction
 F: $[\text{Cr}(\text{OH})_6]^{3-}(\text{aq})$ or $[\text{Cr}(\text{OH})_4]^{-}(\text{aq})$
 G: $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$

(ii) The ability to display variable oxidation states in their compounds. This is due to the

close similarity in energy between the 3d and 4s electrons, which thus allows for different number of these electrons to participate in chemical bonding.

OR Ability to form complexes due to availability of low lying vacant orbitals.

(c) (i)



- (ii) In an octahedral environment, lone pairs on the ligands approach the central metal ion along the x, y and z axes.

$3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals have their greatest electron density along the co-ordinate axes on which the ligands are situated. Hence electrons in these orbitals are pointing towards the lone pairs of ligands, and will be repelled by them.

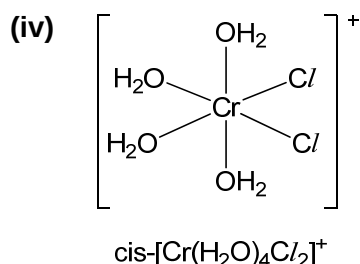
$3d_{xy}$, $3d_{yz}$, $3d_{xz}$ orbitals have their greatest electron density in between the co-ordinate axes. Hence the repulsion between electrons in these orbitals and those of the approaching ligands will be less compared to electrons in $3d_{x^2-y^2}$ or $3d_{z^2}$ orbitals.

Hence the d-orbitals are split into two different energy levels.

- (d) (i) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+$
 $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]^{2+}$
 $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

- (ii) The presence of ligands causes the splitting of the five 3d orbitals in Cr^{3+} into two sets of slightly different energy levels. Since the 3d subshell in Cr^{3+} are partially filled, electrons from the lower-energy d orbitals can absorb energy and get promoted to the higher-energy d orbitals (d-d transition). The energy absorbed corresponds to certain wavelength from the visible light spectrum and the colour observed is the complement of the colour absorbed.

- (iii) Chloride and water are ligands of different strength. Different proportions of these ligands cause the d orbitals to be split to different extents, creating different energy gaps for different complexes, which in turn absorb energies of different wavelengths from the visible light spectrum, thus displaying different colours.



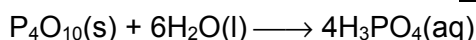
- (e) (i) Positive ($\Delta S > 0$). There is increase in disorder/entropy due to increase in number of particles / molecules / species formed.
- (ii) ΔH value of the forward reaction is close to zero as the type (N→Cr coordinate bond) and the number of dative covalent bonds broken and formed are about the same.
- (iii) $\Delta G = \Delta H - T\Delta S$
Since ΔH is close to zero and ΔS is positive, ΔG (for the forward reaction) will be more negative at a higher temperature, hence increasing the yield of complex **H**.

Section B

- 4 (a) (i) Na_2O reacts with water to form a strongly alkaline solution ($\text{pH} \approx 13 - 14$).
 $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \longrightarrow 2\text{NaOH}(\text{aq})$

SiO_2 does not dissolve in water because of the strong Si–O covalent bonds in the giant molecular structure. Solution remains neutral ($\text{pH} = 7$).

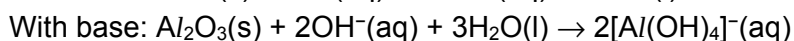
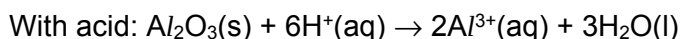
P_4O_{10} reacts with water to form an acidic solution ($\text{pH} \approx 2$).



(Note: H_3PO_4 is a weak acid)

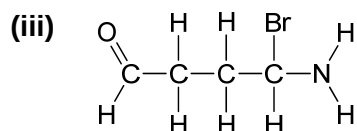
- (ii) (MgO is an ionic oxide which is basic and react with acids. SO_3 is a covalent oxide which is acidic and reacts with bases.)

Al_2O_3 (an amphoteric oxide) reacts with both acids and bases as it is an ionic oxide with covalent character. The covalent character is due to the high charge density/polarising power of Al^{3+} that polarise and distort the electron cloud of O^{2-} .

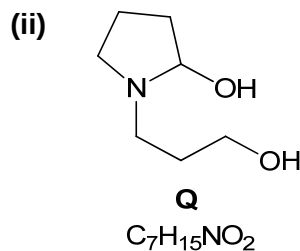
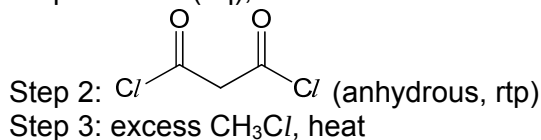


- (b) (i) Step 1: nucleophilic addition

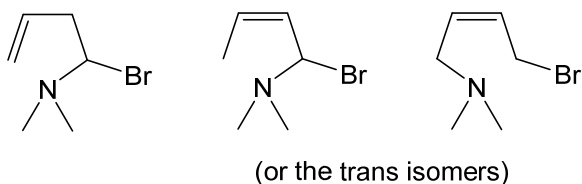
- (ii) In primary amides, the lone pair of electrons on the nitrogen atom interacts with the π electron cloud of the adjacent C=O bond and is delocalised. Hence, the lone pair of electrons is not available for donation to the electrophilic carbonyl carbon atom /unable to act as a nucleophile in step 1.



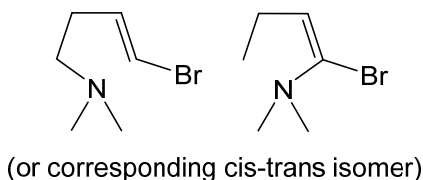
- (c) (i) Step 1: $\text{NaOH}(\text{aq})$, heat



(iii) Possible structures for **S** ($C_6H_{12}BrN$):



Possible structures for **T** ($C_6H_{12}BrN$):



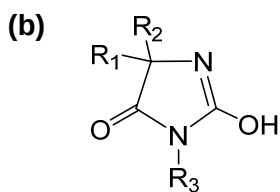
S and **T** are constitutional / structural / positional isomers.

(iv) In **T**, the p orbital of Br overlaps with the π electron cloud of the adjacent $C=C$ bond and the lone pair of electrons in the p orbital of Br delocalises into the $C=C$ bond and results in partial double bond character in the $C-Br$ bond.

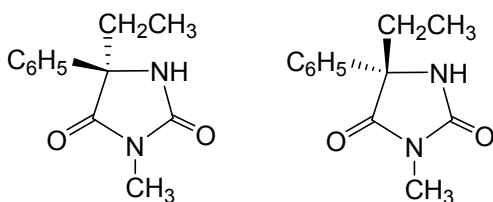
Hence, **T** does not undergo nucleophilic substitution and no Br^- are present to react with Ag^+ to form pale cream ppt of $AgBr$.

(**S** is an alkyl bromide that can undergo nucleophilic substitution when heated with ethanolic $AgNO_3$ to form alcohol and Br^- . Br^- then forms the pale cream ppt of $AgBr$ with Ag^+ .)

5 (a) Constitutional / Structural / Functional group isomerism



(c) (i) Enantiomerism



- (ii) Test: To separate samples of **X** and phenytoin, add NaOH(aq) and heat strongly.

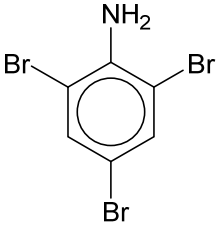
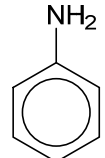
Observation: Pungent gas (NH₃) that turns moist red litmus blue is evolved for the sample containing phenytoin but no pungent gas is evolved for the sample containing **X**.

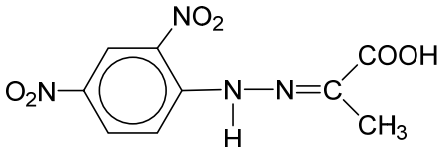
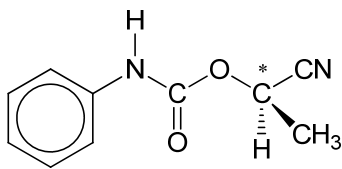
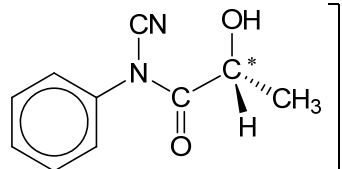
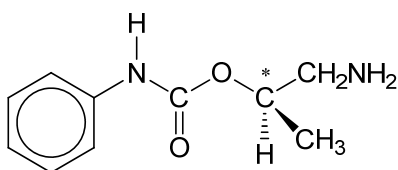
- (d) (i) $M_r(\mathbf{J}) = 190$
 $n(\mathbf{J}) = 5/190 = 0.02632 \text{ mol}$; $n(\text{H}_2) = 2(0.02632) = 0.05264 \text{ mol}$

$$pV_{\text{H}_2} = nRT$$

$$(1.0 \times 10^5) V_{\text{H}_2} = (0.05264)(8.31)(273 + 150)$$

$$V_{\text{H}_2} = 1.85 \times 10^{-3} \text{ m}^3 = 1.85 \text{ dm}^3$$

Information / Reaction	Deduction / Explanation
J has molecular formula, $\text{C}_{10}\text{H}_{10}\text{N}_2\text{O}_2$	J is likely to contain a benzene ring due to the relatively high C to H ratio.
J exhibits optical activity	J contains a chiral centre and/or no plane of symmetry.
J is neutral.	J contains amide and/or $-\text{CN}$ group. Or J is not carboxylic acid or amine or phenol.
J has no reaction with 2,4-DNPH	J is not aldehyde & ketone / carbonyl
$\mathbf{J} \xrightarrow[150^\circ\text{C}]{\text{H}_2(\text{g}) / \text{Ni}} \mathbf{K}$	Reduction of nitrile. Since $\text{C}=\text{C}$ is absent and there is increase of 4 H atoms from J to K , J contains $-\text{CN}$ group.
$\mathbf{J} \xrightarrow[\text{heat}]{\text{H}_2\text{SO}_4(\text{aq})} \text{CO}_2 + \text{resultant mixture}$ distillation $\downarrow$ residue + distillate L residue $\xrightarrow{\text{base}} \mathbf{M}$	J contains amide/ester/nitrile and undergoes acidic hydrolysis. Residue is an ionic salt. (M is a base.)
$\mathbf{M} \xrightarrow{\text{Br}_2(\text{aq})} \mathbf{N} (\text{C}_6\text{H}_4\text{NBr}_3)$	M undergoes electrophilic substitution.  N is  Hence M is

L $\xrightarrow{\text{hot alkaline I}_2(\text{aq})}$ yellow ppt L decolourises acidified $\text{KMnO}_4(\text{aq})$ to form P P $\xrightarrow{2,4\text{-DNPH}}$ Q (orange ppt)	$\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$ L contains the group L undergoes oxidation to form P. Hence, P is a methyl ketone / carbonyl compound. (L cannot be a methyl ketone since J is not carbonyl compound and L is oxidised by KMnO_4 to form P.) $\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{COOH} \\ \\ \text{OH} \end{array}$ Hence, L ($\text{C}_3\text{H}_6\text{O}_3$) is $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{COOH} \end{array}$ P is <u>Condensation</u>  Q is
Combining the above deductions, J can be:  [or]  K can be:  [or] 